

1.7 Human Computer Interaction Problems

ایچ سی آئی مسائل وہ رکاوٹیں ہیں جو سسٹم کو انسانی ضرورت یا ذہنی حد کے مطابق نہ ہونے پر پیدا ہوتی ہیں۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 10 of 15

★ Golden

1.7 Human Computer Interaction Problems ★ Golden

LEARN IT

HCI problems are common difficulties that stop users from using technology smoothly. They happen when systems fail to meet human needs or cognitive limits. **Cognitive limits** means the mind can only hold and process so much at once — if an app asks you to remember three SMS codes at the same time, it has gone past that limit.

Board answers score well when you name the problem, explain it in one line, and give a concrete example (the source examples are safe; a matching local example is even better).

Problem	Explanation and example
Interface complexity	Too many options or a confusing layout. Example: a shopping app with so many menus you cannot find the checkout button.
Poor feedback	The system does not show if an action worked or failed. Example: clicking Submit and the screen freezes with no confirmation.
Inconsistency	Different parts of the app look or act differently. Example: a magnifying-glass icon for Search on one page, a binocular icon on another.
Accessibility gaps	Design ignores people with disabilities. Example: a video tutorial platform without captions for deaf users.
Compatibility issues	Interface fails on different devices or browsers. Example: a website that works on a laptop but crashes on a mobile browser.
Contextual neglect	The system does not adjust to the user's situation. Example: a map app with no dark mode, blinding the driver at night.

High cognitive load	Users must memorize too much or follow overly complex steps. Example: a registration system forcing you to remember three different SMS codes at once.
Trust deficiency	The system feels unsafe or hides security cues. Example: an online store asking for credit cards but lacking an SSL/HTTPS lock icon.

WORKED EXAMPLE

Worked example: diagnose three local systems.

ATM: six tiny unmarked buttons beside the screen and no "Cash is coming" message → interface complexity + poor feedback. A user walks away while cash is still coming out.

JazzCash: "Send" is green on the home screen but orange on the confirm screen, and there is no padlock on the payment page → inconsistency + trust deficiency.

School portal: works in Chrome on a lab PC, blank on a student's Android browser; exam instructions are a caption-free video; login needs a portal password, an email OTP and an SMS OTP on the same screen → compatibility, accessibility gap, high cognitive load.

Once you can name the problem, Section 1.8 tells you how to fix it.

REMEMBER IT

Eight problems in two families: (1) Screen problems — complexity, inconsistency, accessibility, compatibility, context. (2) Mind-and-trust problems — poor feedback, cognitive load, trust. If the user is lost, look at the screen family. If the user is scared or exhausted, look at the mind family.

COMMON MISTAKE

Mistake: Writing "the app is bad" as an HCI problem, or mixing up cognitive load with compatibility.

Fix: Use the exact names from the table. Cognitive load is "too much to remember". Compatibility is "breaks on another device or browser".

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EXAM TIP

Common 8-mark question: "Describe problems of HCI with examples." Aim for at least six of the eight, each in the pattern Name → one-line meaning → example. Memorise the source examples (checkout buried in menus, frozen Submit, binocular vs magnifying glass, no captions, laptop-only site, no dark mode, three SMS codes, missing HTTPS lock). Adding ATM / JazzCash / portal examples shows application and often gains the last marks.

CHECK YOURSELF

1. Which problem is a missing HTTPS lock on a payment page?
2. Why are three SMS codes at once an HCI problem?
3. A site works on a laptop but crashes on a phone. Name the problem.

Answers: 1. Trust deficiency. 2. High cognitive load — users must memorize too much. 3. Compatibility issues.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. HCI studies:

- a. How people and computers work together
- b. Only CPU clocks
- c. Only NAND
- d. Only SQL keys

MCQ 2. Another name mentioned is:

- a. CHI / MMI
- b. MVP
- c. SDLC only
- d. K-map

MCQ 3. Sensory channels include:

- a. Sight, touch, hearing, voice, spatial
- b. Only smell in this course
- c. Only taste
- d. Only OSI layer 7

MCQ 4. Good HCI makes systems:

- a. Usable, efficient, fitting human needs
- b. Harder on purpose
- c. Mouse-only always
- d. Text in 4pt always

MCQ 5. Haptics are:

- a. Touch feedback (vibration)
- b. Pie charts
- c. Foreign keys
- d. Sprints

MCQ 6. VR tracking is a:

- a. Spatial channel
- b. Primary key
- c. Bubble sort
- d. Histogram

Extra short questions

1. Define HCI.
2. Five sensory channels from the notes.
3. Give a computer response for touch and for voice.
4. Why HCI is not "just drawing a pretty screen".

Extra long question

1. Tabulate the five channels with a human action and a computer response. Then explain one device that uses three channels at once.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Human-Computer Interaction: design of technology around people.
2. Sight, touch, hearing, voice, spatial.
3. Touch: click/haptic. Voice: speech recognition and spoken reply.
4. It includes users, tasks, environment, feedback, accessibility.

Long

1. Smartphone: sight (GUI), touch (tap/haptic), hearing (ringtone), optional voice assistant. Closing: if one channel fails, another should still work (accessibility).

Next lecture: 1.8 Methods for Improving HCI